MIL-T-4748B(USAF)
31 August 1981
SUPERSEDING
MIL-T-4748A(USAF)
28 July 1966

MILITARY SPECIFICATION

TESTER, TEMPERATURE INSTRUMENTS, MJ-1

This specification is approved for use by HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of temperature instrument tester, designated Type MJ-1.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited).
QQ-S-741	Steel, Carbon: Structural Shapes, Plates, and Bars.

MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-S-6872	Soldering Process, General Specification For.
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base.

STANDARDS

FEDERAL

FED-STD-595	Colors.

MILITARY

MIL-STD-129	Marking For Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/MMEDO, Kelly Air Force Base, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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MIL-STD-831	Test Reports, Preparation of.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.1 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 366	Steel Sheets, Carbon, Cold Rolled, Commercial Quality.
ASTM A 569	Steel, Carbon (0.15 Maximum Percent), Hot Rolled Sheet and Strip, Commercial Quality.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification provides for preproduction testing.

3.2 Components. The tester shall consist of a single integral unit incorporating mechanical refrigeration and electrical heating system, a liquid bath container, electronic temperature controller and the necessary piping, electrical wiring, and controls to provide a compact light weight facility for the testing and calibration of aircraft temperature instruments.

3.3 Selection of specifications and standards. Specifications and standards for necessary commoditites and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they have suitable properties and are replaceable by AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.4 Materials.

3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, slat spray, or atmospheric conditions likely to be met in storage or normal service.

*3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.4.2 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a suitable fungicidal agent, however, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.5 Design and construction. The tester shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The tester shall be designed for use in the testing and calibration of aircraft temperature instruments in the range of -55° to $+120^{\circ}$ Centigrade (C).

3.5.1 General configuration. The general configuration of the tester shall conform substantially to Figure 1. All control valves, selector switches and gages shall be readily accessible to the operator.

3.5.2 Cabinet. The structure of the tester cabinet shall be fabricated of steel conforming to QQ-S-741. The exterior covering of the cabinet, except screened areas required for ventilation, shall be of sheet steel conforming to ASTM A 366 or ASTM A 569. The cabinet shall be provided with either, or both, adequate doors and easily removable panels to permit access to operating parts for servicing or adjustment purposes.

3.5.2.1 Lifting handles. The cabinet shall be provided with folding type spring handles located on each side to facilitate lifting and moving operations.

3.5.2.2 Operator protection. The cabinet shall be fully guarded to shield belts, pulleys, electrical connections and any other component parts which may present a hazard to the operator.

3.5.3 Mechanical and electrical design. The detailed mechanical and electrical design of this equipment shall be accomplished by the contractor subject to the requirements of this specification, these requirements being detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence in accuracy of indications. The design layout and assembly of the tester and its component parts shall be such as to facilitate quantity production and ease of maintenance and operation.

3.5.4 Electrical power. The tester shall be suitable for operation on a 110 volts, 1 phase, 60 cycle alternating current power source. The external power source shall not be furnished with the tester. Component parts of the tester requiring other potentials, shall obtain their power by means of transformers and rectifiers which shall be installed as a part of the tester. The tester shall incorporate an approved three-prong plug-in type receptacle (third prong to be connected to chassis), conveniently located and easily accessible, for use in connecting the tester to a grounded external power source. The receptacle shall be suitably marked to indicate the electric current required for operation of the tester.

3.5.5 Refrigerant. The refrigeration system of the tester shall be designed and constructed for operation on either, or both, Dichlorodifluoromethane (Freon 12) and Monochlorodifluoromethane (Freon 22) refrigerant.

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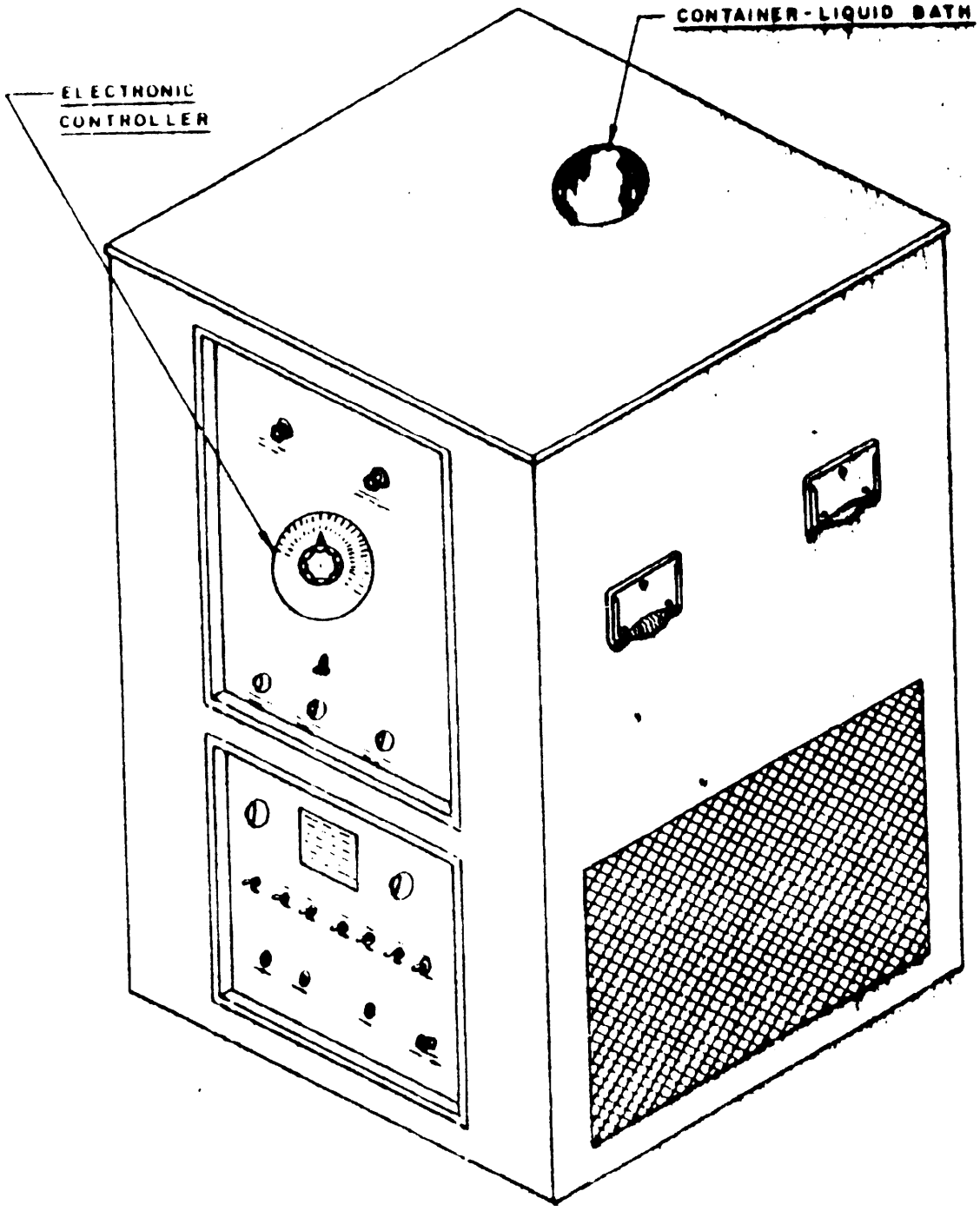


FIGURE 1

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3.5.5.1 Tubing. Refrigerant tubing connections shall be silver soldered or brazed. All refrigerant, oil and water lines shall be properly anchored and provided with flexible connections or loops wherever possible to prevent either, or both, excessive vibration and failure during operation of the tester. All refrigerant lines, and connections on which excessive condensation occurs during operation of the tester shall be properly insulated.

3.5.6 Cooling. The tester shall be designed to operate primarily as an air cooled unit at ambient room temperature below +20°C. Provisions shall be made for water cooling of the essential refrigeration system components to insure satisfactory operation of the tester at ambient room temperature above +20°C. Provisions shall also be made for removing the excessive heat stored in the liquid test container, after completion of a high temperature cycle, in order that the refrigeration cycle can be turned on. The water inlet and water outlet lines shall terminate in suitable valves and fittings to provide service connections. The water connections shall be identified "INLET" and "OUTLET".

3.5.7 Heating. Heating of the liquid bath shall be effected by a suitable electrical system. The heating units shall be easily removable for replacement purposes. The heating rate shall be controllable through the medium of a variable rate heat switch. A pilot light shall be provided to indicate when the heat cycle is in operation.

3.5.8 Test well. The tester shall incorporate one liquid bath test well, with a free working space of not less than 3.5 inches in diameter, and 9 inches in depth. The working space shall be shielded from any agitators or circulating pumps required in order to prevent damage to instruments under test. The test well shall be provided with a suitable drain line and valve to permit removal of the liquid. The top of the test well shall be constructed in such a manner that any liquid spilled in the immediate area surrounding the well will drain back into the well. The well shall be constructed to permit the liquid to be filled within 1 inch of the top working surface of the tester. A suitable lid, which is easily removable, shall be provided to cover the well when the tester is not in operation.

3.5.9 Liquid bath. The liquid bath or heat transfer agent shall consist of a single liquid suitable for use throughout the range of -55° to +120°C. The liquid shall be non-toxic, nonflammable, and non-corrosive.

3.5.10 Controller. The tester shall incorporate an electronic type temperature controller. The controller shall be suitable for use in manually selecting and automatically controlling any desired temperature within the range of -55° to +120°C. The controller shall be suitable for maintaining any selected temperature within the range of -55° to 120°C, with a temperature differential not exceeding ±1°C of the sensing element. The controller shall be graduated in degrees centigrade, designating each 10° graduation.

3.5.11 Panel. The tester shall incorporate a suitable control panel, located in a vertical position on the front side of the cabinet. The panel shall contain

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all the necessary controls, gages, switches, and pilot lights required in operation of the tester. All controls and switches shall be properly marked to indicate their function.

3.5.12 Diagrams. One wiring diagram and one piping diagram, permanent in nautre and as large in size as practicable, shall be mounted on the inside of one of the tester panels or doors. The diagrams shall contain pertinent information on the electrical systems and the piping systems of the tester to which reference can be made during servicing operations.

3.5.13 Gages. The tester shall be furnsihed with suitable gages, mounted on the control panel, to provide indications for the pressure in the refrigeration system.

3.5.14 Accessories. The following accessories shall be furnished with each tester:

a. One grounded three-conductor extension power cable, for connecting the tester to the power source. Connection of the power cable to a power source receptacle must be with a grounded third conductor.
b. Two quarts of liquid for the temperature bath.
c. Necessary tools and freon charging tube.

3.5.15 Glides. The tester shall be mounted on four rubber cushioned glides, with provisions incorporated for leveling the cabinet.

3.6 Performance. The tester shall be capable of satisfactorily meeting the following requirements:

3.6.1 Low temperature. The completely assembled tester shall be capable of decreasing the temperature of the liquid bath from +25°C to -55°C, at standard atmospheric pressure in not more than 2 hours.

3.6.2 High temperature. The completely assembled tester shall be capable of increasing the temperature of the liquid bath from +25°C to 120°C, at standard atmospheric pressure in not more than 30 minutes.

3.6.3 Accuracy. The completely assembled tester shall be capable of producing and maintaining any selected temperature within the range specified herein, within an accuracy of ±1°C. Temperature variations within the liquid bath shall not exceed ±.5°C at any point when measurements are taken simultaneously.

*3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.8 Dimensions. The overall dimensions of the completely assembled tester shall not exceed 18 inches in width, 18 inches in depth, and 28 inches in height.

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*1.1 This specification covers two types of lithium

3.9 Weight. The weight of the completely assembled tester shall not exceed 250 pounds.

3.10 Finishes and protective coatings.

3.10.1 External metal surfaces. The external metal surfaces of the completely assembled tester shall be finished in accordance with MIL-STD-808. The color shall be No. 36118 of FED-STD-595. All engraved lines, numerals and lettering shall be filled with white enamel.

3.10.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with Specification QQ-P-416, Type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship.

3.12.1 General. The tester including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts, etc.

3.12.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the threads or attaching parts.

3.12.3 Cleaning. The tester shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, and other foreign material will be removed during and after final assembly.

3.12.4 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

3.13 Reclaimed material. The use of reclaimed material shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of testers shall be classified as follows:

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a. Preproduction tests.
b. Acceptance tests.

4.3 Test Conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.4 Preproduction testing.

4.4.1 Test sample. The test sample shall consist of one tester representative of the production equipment. The sample shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity (see 6.2c).

4.4.2 Test report. Upon completion of preproduction testing, the contractor shall submit a test report prepared in accordance with MIL-STD-831.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests under "test methods" in 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of all tests under "test methods" in 4.6.

4.6 Test methods.

4.6.1 Examination of product. Each tester shall be thoroughly inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Low temperature operation. The tester, with the test well filled with the liquid agent and the refrigeration system operating, shall be operated for a period of 2 hours, after which time the temperature of the liquid bath agent shall be $-55 \pm 2^{\circ}C$. While being maintained at this temperature the tester shall meet the requirements specified in 3.6.3.

4.6.3 High temperature operation. The tester, with the test well filled with the liquid agent and the heating system operating shall be operated for a period of 30 minutes, after which time the temperature of the liquid agent shall be $+120 \pm 2^{\circ}C$. While being maintained at this temperature the tester shall meet the requirements specified in 3.6.3.

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storgae shall be in accordance with the requirements of section 5, or the documents specified therein.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each tester shall be preserved and packaged in accordance with Method IIA of MIL-P-116. The unit container of the Method IIA package shall conform to MIL-B-26195. Refer to paragraphs 5.2.1 and 5.2.2 for the appropriate class, style and type necessary for the specified level of packing.

5.1.2 Level C. Each tester shall be preserved and packaged to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the initial receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. The unit container shall conform to MIL-B-26195, Class 2, Style A, Type I. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.2 Level B. Level B packing shall be the same as Level A except that the container shall be Class 1, Style A, Type I.

5.2.3 Level C. Each tester preserved and packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe transportation to the first receiving activity. Containers shall conform to Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Tests: Edgewise drop test or cornerwise drop test shall be in accordance with Appendix A of the standard. Tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MJ-1 tester covered by this specification is intended to be used for testing and calibrating aircraft temperature instruments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

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b. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

c. Instructions for preproduction sample (see 4.4.1).

6.3 Identification of changes. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

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